

Dynamic Macroeconomics II

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Exogenous Growth in the Neoclassical Model: Detrending

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Exogenous sources of growth and computational issues

- Suppose we model an economy in which either productivity and/or population grow
- Assume Total Factor Productivity and population grow at exogenous rates.
- Therefore (some endogenous) variables exhibit growth in long run; a steady state (in levels) does not exist.
- Shooting algorithm and log-linearization techniques (that we will see soon) cannot solve these models unless we **detrend** them.
- Those techniques require steady state.
- Dynamic programming requires a maximum capital, but with exogenous growth it is unbounded.
- Solution: transform model to remove growth trend of (some) variables.
- Then we can apply techniques we know.
- To present these ideas we use McGrattan (2006), without uncertainty.

- N_t : total population, $N_0 = 1$, $N_t = N_0(1 + g_n)^t$.
- Z_t : labor-augmenting technological change. $Y_t = F(K_t, Z_t L_t)$, $Z_t = (1 + g_z)^t$.
- Denote by V_t the aggregate value of a variable, v_t its per-capita counterpart and $\widehat{v}_t$ the detrended value of a variable. That is:

$$v_t = \frac{V_t}{N_t}$$
$$\widehat{v}_t = \frac{V_t}{N_t(1 + g_z)^t}$$

- Utility depends on consumption and leisure $1 - l_t$.
- Given prices w_t, r_t , tax rates $\tau_t^c, \tau_t^x, \tau_t^k, \tau_t^l$, and government transfers tr_t , household solves:

$$\max_{\{c_t, x_t, l_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t [U(c_t, 1 - l_t)] N_t$$

s.t.

$$\begin{aligned} (1 + \tau_t^c)c_t + (1 + \tau_t^x)x_t &= (1 - \tau_t^k)r_t k_t + (1 - \tau_t^l)w_t l_t + \tau_t^k \delta k_t + tr_t \\ N_{t+1}k_{t+1} &= [(1 - \delta)k_t + x_t]N_t \\ c_t, x_t &\geq 0 \end{aligned}$$

How did we rewrite the capital accumulation equation?

$$K_{t+1} = (1 - \delta)K_t + X_t$$
$$\frac{K_{t+1}}{N_t} \frac{N_{t+1}}{N_{t+1}} = (1 - \delta) \frac{K_t}{N_t} + \frac{X_t}{N_t}$$

which can be rewritten for example as

$$(1 + g_n)k_{t+1} = (1 - \delta)k_t + x_t$$

or (equivalently) as in the previous slide.

- Firm solves:

$$\max_{\{K_t, L_t\}} F(K_t, Z_t L_t) - w_t L_t - r_t K_t$$

- Government's budget constraint (note that public expenditure G_t is exogenous):

$$G_t + N_t t r_t = \tau_t^k (r_t - \delta) N_t k_t + \tau_t^l w_t l_t N_t + \tau_t^c N_t c_t + \tau_t^x N_t x_t$$

- In equilibrium, the following feasibility conditions must hold:

$$N_t (c_t + x_t) + G_t = F(K_t, Z_t L_t)$$

$$N_t k_t = K_t$$

$$N_t l_t = L_t$$

The Neoclassical Model is designed to produce outcomes that match **long-run** stylized facts of developed economies, notably the United States and Great Britain. These are (some of) **Kaldor's facts**. Among them:

- GDP per capita, consumption per capita, investment per capita, the real wage, all grow at the same rate.
- Labor per worker and the real interest rate do not have a trend, but a stationary value.
- Therefore in the model we will detrend variables that, by construction, display long-run growth.
- As mentioned earlier, this detrended model can be solved in the computer with methods we know.

- Assume household's utility is given by consumption and decreasing function v of labor (bad notation, already used v):

$$U(c_t, 1 - l_t) = \frac{c_t^{1-\sigma}}{1-\sigma} v(l_t)$$

- Solve for c_t from definition of $\hat{c}_t$:

$$c_t = (1 + g_z)^t \hat{c}_t$$

- Then substitute in utility; also substitute N_t for $(1 + g_n)^t$
- After doing some algebra (**check**):

$$\sum_{t=0}^{\infty} (\beta(1 + g_n)(1 + g_z)^{1-\sigma})^t \frac{\hat{c}_t^{1-\sigma}}{1-\sigma} v(l_t)$$

- Define the modified discount factor as:

$$\tilde{\beta} = \beta(1 + g_n)(1 + g_z)^{1-\sigma}$$

- Therefore, write the intertemporal utility as:

$$\sum_{t=0}^{\infty} \tilde{\beta}^t \frac{\hat{c}_t^{1-\sigma}}{1-\sigma} v(l_t), \text{ need } \tilde{\beta} < 1$$

To work with budget constraint, we do as before, substituting (all but labor) per capita for detrended variables.

- Household's detrended budget constraint (**check**):

$$(1 + \tau_t^c)\widehat{c}_t + (1 + \tau_t^x)\widehat{x}_t = (1 - \tau_t^k)r_t\widehat{k}_t + (1 - \tau_t^l)\widehat{w}_t l_t + \tau_t^k \delta \widehat{k}_t + \widehat{tr}_t$$

- Detrended capital accumulation (**check**):

$$(1 + g_z)(1 + g_n)\widehat{k}_{t+1} = (1 - \delta)\widehat{k}_t + \widehat{x}_t$$

- For the firm's problem, suppose F is homogeneous of degree 1 and transform the production function:

$$F\left(\frac{K_t}{N_t Z_t}, \frac{Z_t L_t}{N_t Z_t}\right) = \frac{F(K_t, Z_t L_t)}{N_t Z_t} \equiv f(\widehat{k}_t, l_t)$$

- Rewrite the firm's objective function:

$$\begin{aligned} \frac{F(K_t, Z_t L_t)}{N_t Z_t} - \frac{w_t L_t}{N_t Z_t} - \frac{r_t K_t}{N_t Z_t} \\ = f(\widehat{k}_t, l_t) - \widehat{w}_t l_t - r_t \widehat{k}_t \end{aligned}$$

- Modify feasibility :

$$\begin{aligned} \frac{Y_t}{N_t Z_t} &= \frac{C_t + X_t + G_t}{N_t Z_t} \\ \implies \widehat{y}_t &= \widehat{c}_t + \widehat{x}_t + \widehat{g}_t \end{aligned}$$

- In the long run, assume g_t grows at a rate g_z , so that $\widehat{g}_t$ grows at rate 0.

- Lagrangian for the household's optimization problem:

$$\mathcal{L} = \sum_{t=0}^{\infty} \left\{ \tilde{\beta}^t U(\widehat{c}_t, 1-l_t) + \lambda_t [(1-\tau_t^k) r_t \widehat{k}_t + (1-\tau_t^l) \widehat{w}_t l_t + \tau_t^k \delta \widehat{k}_t + \widehat{t} r_t - (1+\tau_t^c) \widehat{c}_t - (1+\tau_t^x) \widehat{x}_t] + \mu_t [(1-\delta) \widehat{k}_t + \widehat{x}_t - (1+g_z)(1+g_n) \widehat{k}_{t+1}] \right\}$$

- Solving the above problem, we obtain the static consumption-leisure condition:

$$\frac{U_2(\widehat{c}_t, 1-l_t)}{U_1(\widehat{c}_t, 1-l_t)} = \frac{1-\tau_t^l}{1+\tau_t^c} \widehat{w}_t$$

Euler equation requires an extra step compared to the usual. Take derivative with respect to investment $\widehat{x}_t$. That gives us a relation between multipliers λ_t and μ_t . Take derivative with respect to k_{t+1} , use the previous relation to get (**check**)



$$\frac{1 + \tau_t^x}{1 + \tau_t^c} U_1(\widehat{c}_t, 1 - l_t) = \widehat{\beta} \left\{ \frac{U_1(\widehat{c}_{t+1}, 1 - l_{t+1})}{1 + \tau_{t+1}^c} [(1 - \tau_{t+1}^k) r_{t+1} + \delta \tau_{t+1}^k + (1 - \delta)(1 + \tau_{t+1}^x)] \right\}$$

where

$$\widehat{\beta} = \frac{\widetilde{\beta}}{(1 + g_z)(1 + g_n)} = \beta(1 + g_z)^{-\sigma}$$

- First-order conditions for firm's problem:

$$r_t = f_1(\widehat{k}_t, l_t)$$

$$\widehat{w}_t = f_2(\widehat{k}_t, l_t)$$

We are ready to use computational methods

- Plug prices from firm's problem into Euler and Consumption-leisure.
- Plug Capital accumulation equation into Feasibility.
- We now have a system of 3 unknowns: $\hat{c}_t, l_t, \hat{k}_{t+1}$.
- We have 3 equations: Euler, Consumption-leisure, and Feasibility.
- Once every variable has been detrended, the usual methods for solving dynamic models can be applied.
- We can recover $\hat{x}_t$ from the solution.
- Done.